INCIDENT REPORT SSH FAILED LOGIN ANALYSIS

Log File: auth.log
Incident Type: Suspected SSH Brute-Force / Unauthorized Login Attempt
Primary Suspicious IP: 203.0.113.50
Failed Login Attempts: 4
Successful Login: Yes – root account

1. Incident Summary

The analysis of auth.log identified multiple failed SSH authentication attempts from several IP addresses. The most suspicious activity originated from 203.0.113.50, which generated four consecutive failed login attempts against the root account and subsequently achieved a successful root login.

2. Suspicious IP Address

Primary suspicious IP: 203.0.113.50

IP Address	Failed Attempts
192.168.1.10	3
10.0.0.5	3
203.0.113.50	4

3. Number of Attempts

Total failed SSH login attempts: 10. Breakdown: 192.168.1.10 = 3; 10.0.0.5 = 3; 203.0.113.50 = 4.

4. Evidence and Commands Used

4.1 . View the complete log

```
cat auth.log
```

Purpose: Display the contents of the authentication log.

4.2 . View the end of the log

```
tail auth.log
```

Purpose: Review the most recent entries.

4.3 . Identify failed SSH login attempts

```
grep "Failed password" auth.log
```

Purpose: Extract failed SSH password-authentication events.

4.4 . Count total failed login attempts

```
grep "Failed password" auth.log | wc -l
```

Purpose: Result: 10 failed login attempts.

4.5 . Extract source IP addresses

```
grep "Failed password" auth.log | awk '{print $11}'
```

Purpose: Extract the source IP address from each failed-login entry.

4.6 . Count failed attempts by IP

```
grep "Failed password" auth.log | awk '{print $11}' | sort | uniq -c
```

Purpose: Result: 3 for 192.168.1.10; 3 for 10.0.0.5; 4 for 203.0.113.50.

4.7 . Sort IP addresses by attempts

```
grep "Failed password" auth.log | awk '{print $11}' | sort | uniq -c | sort -nr
```

Purpose: Places the IP with the highest number of failures first.

4. 8 . Identify IPs with more than three failures

```
grep "Failed password" auth.log | awk '{print $11}' | sort | uniq -c | awk '$1 > 3'
```

Purpose: Result: 4 for 203.0.113.50.

4.9 . Identify successful logins

```
grep "Accepted password" auth.log
```

Purpose: Find successful SSH password-authentication events.

4.10 . Extract successful-login IP addresses

```
grep "Accepted password" auth.log | awk '{print $11}'
```

Purpose: Result: 192.168.1.5 and 203.0.113.50.

4.11 . Monitor failed logins in real time

```
tail -f auth.log | grep "Failed password"
```

Purpose: Continuously display new failed SSH login attempts.

5. Conclusion – Was the System Compromised?

Based on the log, there is evidence of a successful unauthorized authentication and potential system compromise. The evidence shows that 203.0.113.50 attempted to log in as root, failed four times, and successfully authenticated as root. The successful authentication occurred 35 seconds after the last failed attempt.

However, auth.log alone does not prove what actions were performed after the successful login.

Escalate the incident to the next-level security team for detailed investigation of the successful `root` authentication.

6. LEARNING OUTCOME – WHAT THEY REALLY DID

Learned how to investigate SSH authentication logs using basic Linux commands such as `grep`, `awk`, `sort`, `uniq`, `wc`, and `tail`. Identified failed login attempts, counted and grouped attempts by source IP, identified the most suspicious IP, and correlated failed attempts with a subsequent successful login.

The key learning was that **`203.0.113.50` made four failed attempts to access the `root` account and then successfully logged in as `root`**. This demonstrates how log analysis can help identify potential brute-force activity and possible account compromise.

6. **BONUS CHALLENGE**

```
┌──(shockwave㉿shockwave)-[~/Documents]
└─$ tail -f auth.log | grep "Failed password"
Apr 13 09:00:10 server sshd[1023]: Failed password for admin from 192.168.1.10 port 4457 ssh2
Apr 13 09:01:15 server sshd[1024]: Failed password for root from 10.0.0.5 port 2233 ssh2
Apr 13 09:01:20 server sshd[1025]: Failed password for root from 10.0.0.5 port 2234 ssh2
Apr 13 09:01:25 server sshd[1026]: Failed password for root from 10.0.0.5 port 2235 ssh2
Apr 13 09:02:10 server sshd[1028]: Failed password for root from 203.0.113.50 port 7788 ssh2
Apr 13 09:02:15 server sshd[1029]: Failed password for root from 203.0.113.50 port 7789 ssh2
Apr 13 09:02:20 server sshd[1030]: Failed password for root from 203.0.113.50 port 7790 ssh2
Apr 13 09:02:25 server sshd[1031]: Failed password for root from 203.0.113.50 port 7791 ssh2
Apr 13 09:05:00 server sshd[2000]: Failed password
Apr 13 09:05:00 server sshd[2000]: Failed password for root from 45.33.32.1 port 9999 ssh2
Apr 13 09:05:00 server sshd[2000]: Failed password for root from 45.33.32.1 port 9999 ssh2
^C

┌──(shockwave㉿shockwave)-[~/Documents]
└─$ grep "Failed password" auth.log | cut -d ' ' -f 3
09:00:01
09:00:05
09:00:10
09:01:15
09:01:20
09:01:25
09:02:10
09:02:15
09:02:20
09:02:25
09:05:00
09:05:00
09:05:00

┌──(shockwave㉿shockwave)-[~/Documents]
└─$ grep "Failed password" auth.log | awk '{print $11}' | sort | uniq -c | sort -nr | head -3
      4 203.0.113.50
      3 192.168.1.10
      3 10.0.0.5
```